

MATH 118 Homework Assignment 2

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Q1)

1.1) For a discrete probability distribution to be valid it must satisfy two conditions:

1) $0 \leq P(X=x) \leq 1$ for all x

0.1, 0.2, 0.4, 0.2, 0.1 are within this range.

2) Sum of all probabilities must be 1.

$$0.1 + 0.2 + 0.4 + 0.2 + 0.1 = 1.0$$

Therefore, this is a valid probability distribution.

1.2) $E[X]$ or mean of X is weighted average of all possible values, using their probabilities as weights:

$$E[X] = \sum x_i P(X=x_i)$$

$$E[X] = (0)(0.1) + 1(0.2) + 2(0.4) + 3(0.2) + 4(0.1) = 2.0$$

$$E[X] = 2.0$$

Var(X): Variance measures how spread out the values are around the mean or $E[X]$.

$$\text{Var}(X) = E[X^2] - (E[X])^2 \text{ where } E[X^2] = \sum x_i^2 P(X=x_i)$$

$$E[X^2] = (0^2)(0.1) + (1^2)(0.2) + (2^2)(0.4) + (3^2)(0.2) + (4^2)(0.1)$$

$$E[X^2] = 0 + 0.2 + 1.6 + 1.8 + 1.6 = 5.2 \Rightarrow E[X^2] = 5.2$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 5.2 - (2.0)^2 = 1.2$$

$$\text{Var}(X) = 1.2$$

1.3) Find $P(X \geq 2)$

$$P(X \geq 2) = P(X=2) + P(X=3) + P(X=4)$$

$$P(X=2) = 0.4, P(X=3) = 0.2, P(X=4) = 0.1$$

$$P(X \geq 2) = 0.4 + 0.2 + 0.1 = 0.7$$

1.4)

• X counts defect on a single product.

$$E[X] = \sum_x x P(X=x) = 2.0 \text{ defects/product}$$

This is the probability-weighted average of possible defect counts.

• If you examine many products from the same process, the average defects per product will settle around 2.

• For a batch of n products the expected total defect is

$$E[\text{total defects}] = n \cdot E[X] = 2n$$

In 1000 products, expect about 2000 total defects.

• $E[X]=2$ is not saying every product has 2 defects. Single items can have 0, 1, 2, 3, 4 defects.

• The spread matters: $\text{Var}(X) = 1.2$ implies noticeable variability around 2.

• $P(X=0) = 0.1 \Rightarrow$ Only 10% of products are defect-free under this distribution.

• If fixing one defect costs c , the expected rework cost per product is $E[\text{cost}] = c \cdot E[X] = 2c$

On average, the process produces 2 defects per product; hence $2n$ expected defects in n units.

Q2: $f(y) = \begin{cases} k(1-y^2), & -1 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$

2.1: A valid pdf must satisfy:

$$\int_{-\infty}^{\infty} f(y) dy = 1$$

Since $f(y) = 0$ outside $[-1, 1]$

$$\int_{-1}^1 k(1-y^2) dy = 1$$

$$= k \int_{-1}^1 (1-y^2) dy = 1$$

$$k \left[y - \frac{y^3}{3} \right]_{-1}^1 = 1 \Rightarrow k \left[\left(1 - \frac{1}{3}\right) - \left(-1 + \frac{1}{3}\right) \right] = 1 \Rightarrow k \left[\frac{2}{3} + \frac{2}{3} \right] = 1$$

$$\Rightarrow k \frac{4}{3} = 1 \Rightarrow \boxed{k = \frac{3}{4}}$$

2.2:

• For $y < -1$ $F(y) = 0$

• For $-1 \leq y \leq 1$ $F(y) = \int_{-\infty}^y \frac{3}{4}(1-t^2) dt$ $F(y) = \frac{3}{4} \left(t - \frac{t^3}{3} \right)_{-1}^y$

$$F(y) = \frac{3}{4} \left(y - \frac{y^3}{3} + \frac{2}{3} \right)$$

• For $y > 1$ $F(y) = 1$

$$F(y) = \begin{cases} 0 & y < -1 \\ \frac{3}{4} \left(y - \frac{y^3}{3} + \frac{2}{3} \right) & -1 \leq y \leq 1 \\ 1 & y > 1 \end{cases}$$

$F(-1) = 0, F(1) = 1 \rightarrow$ The CDF is continuous and valid.

2.3 $P(-0.5 \leq Y \leq 0.5)$

$$P(a \leq Y \leq b) = F(b) - F(a)$$

where $F(y) = \frac{3}{4} \left(y - \frac{y^3}{3} + \frac{2}{3} \right)$ for $-1 \leq y \leq 1$

$$F(0.5) = \frac{3}{4} \left(0.5 - \frac{(0.5)^3}{3} + \frac{2}{3} \right) = 0.84375$$

$$F(-0.5) = \frac{3}{4} \left(-0.5 - \frac{(-0.5)^3}{3} + \frac{2}{3} \right) = 0.15625$$

$$P(-0.5 \leq Y \leq 0.5) = 0.84375 - 0.15625 = \boxed{0.6875}$$

That means there is a 68.75% probability that Y lies between -0.5 and 0.5 .

2.4 $f(y) = \begin{cases} \frac{3}{4}(1-y^2) & -1 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$

$$E[Y] = \int_{-\infty}^{+\infty} y f(y) dy$$

Since $f(y)$ is symmetric around 0 positive and negative parts cancel:

mean or $E[Y] = 0$

$$E[Y^2] = \int_{-1}^1 y^2 \frac{3}{4}(1-y^2) dy$$

$$E[Y^2] = \frac{3}{4} \int_{-1}^1 (y^2 - y^4) dy \Rightarrow E[Y^2] = \frac{3}{4} \left[\frac{y^3}{3} - \frac{y^5}{5} \right]_{-1}^1 \Rightarrow E[Y^2] = \frac{3}{4} \left(\frac{2}{3} - \frac{2}{5} \right)$$

$$\boxed{E[Y^2] = \frac{3}{4} \times \frac{4}{15} = \frac{1}{5}}$$

Variance: $\text{Var}(Y) = E[Y^2] - (E[Y])^2 = 0.2 - (0)^2 = 0.2$

$$\boxed{\text{Var}(Y) = 0.2}$$

03: $f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
 Define $Y = X^2$

3.18

$$E[X] = \int_{-\infty}^{+\infty} x f_X(x) dx = \int_0^1 x (2x) dx = 2 \int_0^1 x^2 dx = 2 \left[\frac{x^3}{3} \right]_0^1$$

$$E[X] = 2 \cdot \frac{1}{3} = \frac{2}{3}$$

$$E[X^2] = \int_0^1 x^2 f_X(x) dx = \int_0^1 x^2 (2x) dx = 2 \int_0^1 x^3 dx = 2 \left[\frac{x^4}{4} \right]_0^1$$

$$E[X^2] = 2 \cdot \frac{1}{4} = \frac{1}{2}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

$$\text{Var}(X) = \frac{1}{18} \approx 0.0556$$

3.20 $E[Y], \text{Var}[Y]$

$Y = X^2$ $f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$

$$E[Y] = E[X^2] = \int_0^1 x^2 f_X(x) dx = \int_0^1 x^2 (2x) dx = 2 \int_0^1 x^3 dx = 2 \left[\frac{x^4}{4} \right]_0^1$$

$$E[Y] = \frac{1}{2}$$

$$E[Y^2] = E[X^4] = \int_0^1 x^4 (2x) dx = 2 \int_0^1 x^5 dx = 2 \left[\frac{x^6}{6} \right]_0^1 = \frac{1}{3}$$

$$E[Y^2] = \frac{1}{3}$$

$$\text{Var}[Y] = E[Y^2] - (E[Y])^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$\text{Var}[Y] = \frac{1}{12} \approx 0.0833$$

3.3: $E[X] = \frac{2}{3}$, $\text{Var}(X) = \frac{1}{18}$, $E[Y] = E[X^2] = \frac{1}{2}$

$$(E[X])^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9} \approx 0.4444 \quad E[Y] = 0.5$$

$$E[Y] = 0.5 \quad (E[X])^2 = 0.4444$$

$$E[Y] > (E[X])^2$$

From the definition of variance:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Since $\text{Var}(X) > 0$ it always follows that:

$$E[X^2] > (E[X])^2$$

$$E[Y] - (E[X])^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Hence, $E[Y]$ exceeds $(E[X])^2$ by exactly the variance of X .